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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Alt + Win + R
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8; supporting alignment with SDG 9
Project / Solution Name	Tepat Bill
Name (Leader)	Isaac Toh Zhen Yong
Email & Phone (Leader)	isaactoh2610@gmail.com & +60 10-557 6808
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small shops that receive supplier deliveries need to reconcile what was ordered, what arrived and what was billed. Paper delivery notes, photographs and messages can leave these records disconnected. A shortage or wrong item recorded at delivery can be missed when an invoice is checked later. Owners then spend time finding evidence and contacting the supplier, while overpayments reduce cash available for stock and daily operations.

Our team reports one interview with a cafe owner. This is early discovery evidence, not a representative survey. We make no claim about how frequently Malaysian shops experience overbilling.

The economic exposure matters even for small discrepancies. Department of Statistics Malaysia, Economic Census 2023: Wholesale and Retail Trade (reference year 2022), https://www.dosm.gov.my/portal-main/release-content/economic-census-2023-wholesale-and-retail-trade-, reports a retail gross margin of 30.9% in Economic Census 2023, referring to 2022.At that illustrative margin, RM100 of avoidable cost requires approximately RM324 in additional sales to generate the same gross profit, assuming the margin remains constant. This is a calculation, not measured Tepat Bill savings.

Tepat Bill focuses on the receiving staff who record deliveries and the owner/manager who reviews differences. The intended benefits are clearer dispute evidence, less repetitive checking and better control over supplier balances. These benefits still require measured validation.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Tepat Bill is a mobile-first delivery and supplier-invoice checking prototype. Staff upload an order, invoice or delivery note and confirm the actual received quantities. Google Gemini extracts document fields; users can correct the extracted lines before saving. Deterministic TypeScript rules compare invoice items, received quantities and reference prices, then calculate differences in Malaysian ringgit. AI does not

calculate the claim amounts.
The current rules identify shortages, price changes and items not ordered. Differences appear as reviewable flags. Staff record information; the owner approves or dismisses flags and records supplier responses. After approval, a WhatsApp link opens a factual draft and evidence link for the owner to send. This is not automatic messaging or proof that the message was delivered.

The ledger records supplier-accepted credits and redeliveries. Accepted credits reduce an unpaid balance or carry forward after payment. A foreign-supplier self-billing flow provides an owner-reviewed preview only. CSV export supports manual accountant handoff. Tepat Bill is designed as a potential Orbyte module. The prototype demonstrates local staff/owner permissions, approvals and activity records. Production integration with Orbyte remains proposed; the intended integration would map these functions to Orbyte's platform services and pass approved results to connected services through an assumed interface. Direct accounting-partner submission, actual fund transfers and tax filing are not implemented. .

The app uses React, Express, PostgreSQL and Replit deployment. Relevant workshop themes are generative-AI APIs, database design and cloud deployment; Evaluation will use synthetic documents. No current accuracy or field-savings result is claimed.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Browser UI: React 19.3.1, Vite 8.3.5, TypeScript 5.9.3 and Tailwind CSS 4.3.3. Backend: Node.js 20.x configured in Replit, Express 5.3.0 and TypeScript. PostgreSQL is accessed through Drizzle ORM 0.45.4 and pg 8.23.1. Uploaded documents use Replit App Storage with a local-disk fallback.

AI extraction: Google Gemini through Vercel AI SDK 7.0.149 and @ai-sdk/google 4.0.103. The source configures gemini-3.8-flash and gemini-3.5-flash-lite; Zod validates extracted fields, while shared TypeScript rules and the ledger calculate amounts in integer sen.

Flow: mobile browser -> Express API -> document storage and Gemini extraction -> user confirmation-> PostgreSQL records -> deterministic comparison -> owner decision -> WhatsApp draft/evidence page or CSV export.

Replit is the confirmed deployment platform. The code contains no RunPod workload; Claude Code was used for most coding; Codex assisted review and documentation. These are development tools, not the runtime document reader.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The team reports one cafe-owner interview as initial discovery. The prototype translates the proposed workflow into two stages: capture delivery evidence and check the supplier invoice against saved records. Its mobile interface, Malay/English text and separate staff/owner permissions support the intended users. These are design choices, not proof of usability.

The application is deployed on Replit. Independent review found count inconsistencies, a count-to-invoice navigation failure, evidence-linking gaps and calculation edge cases. We fixed these and covered them with automated regression tests (248 tests). Before the final demo we will re-test the full journey on the deployed app and on a phone.

For Phase 2, the proposed validation is a fresh team-authored evaluation set with known quantities and prices, including correct documents as controls. We will record detection, false positives, exact RM amounts and human review time. No completed usability study, production readiness or measured time saving is claimed.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary - SDG 8: Decent Work and Economic Growth** **Target 8.2:** Tepat Bill supports productivity through technological upgrading by automating repetitive delivery and invoice comparisons. **Target 8.3:** It supports micro and small businesses by providing clearer supplier records and protecting SME cash flow by catching billing errors before payment and keeps auditable record **Supporting - SDG 9: Industry, Innovation and Infrastructure** Tepat Bill converts paper delivery notes, invoices and WhatsApp orders into structured digital records. Its proposed reusable Orbyte integration could support supplier coordination across multiple SMEs. This provides an indirect supporting link to **Target 9.3's integration of small enterprises into value chains and markets**. **Sources:** https://sdgs.un.org/goals/goal8 and https://sdgs.un.org/goals/goal9 Sources: Department of Statistics Malaysia, Economic Census 2023: Wholesale and Retail Trade (reference year 2022), https://www.dosm.gov.my/portal-main/release-content/economic-census-2023-wholesale-and-retail-trade-

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	An independent performance baseline has not yet been measured. Our existing demonstration fixture, invoice LB-7781, has an invoice total of RM593.70 and two planted discrepancies worth RM53.90: RM41.40 for missing goods and RM12.50 for a price difference. These discrepancies are already recorded in the seed data, so this fixture illustrates the workflow rather than independently validating detection accuracy. Source: project repository, fixtures/seed.json. Phase 2 evaluation will use fresh synthetic documents with independently prepared ground-truth answers.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of the known overbilling value correctly identified before payment, measured by RM value, together with the false-positive rate on correct invoice lines.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Our proposed target is to identify at least 75% of the known overbilling value across 12 fresh synthetic delivery–invoice cases, comprising eight cases containing planted discrepancies and four correct control cases, with no false flags on correct lines. Another team member will prepare the documents and ground-truth answers. The cases will not be used during development, preloaded as flags, or supplied through fixture answers. This is a proposed evaluation target, not an achieved result or guaranteed customer saving.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

[/] No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React; Vite; Express; Tailwind CSS	Code	MIT
TypeScript; Vercel AI SDK; @ai-sdk/google; Drizzle ORM; PDF.js	Code	Apache-2.0
Google Gemini configured model identifiers	Hosted pretrained models	models Google Gemini API terms; not an Apache-licensed model.
Team synthetic documents and expected JSON	Synthetic evaluation/demo data	Repository Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

[/] No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[24/10/2026]